

5.1.1 LEAP YEAR

ALGORITHM

Step 1: Start

Step 2: Read the year

Step 3: Check if

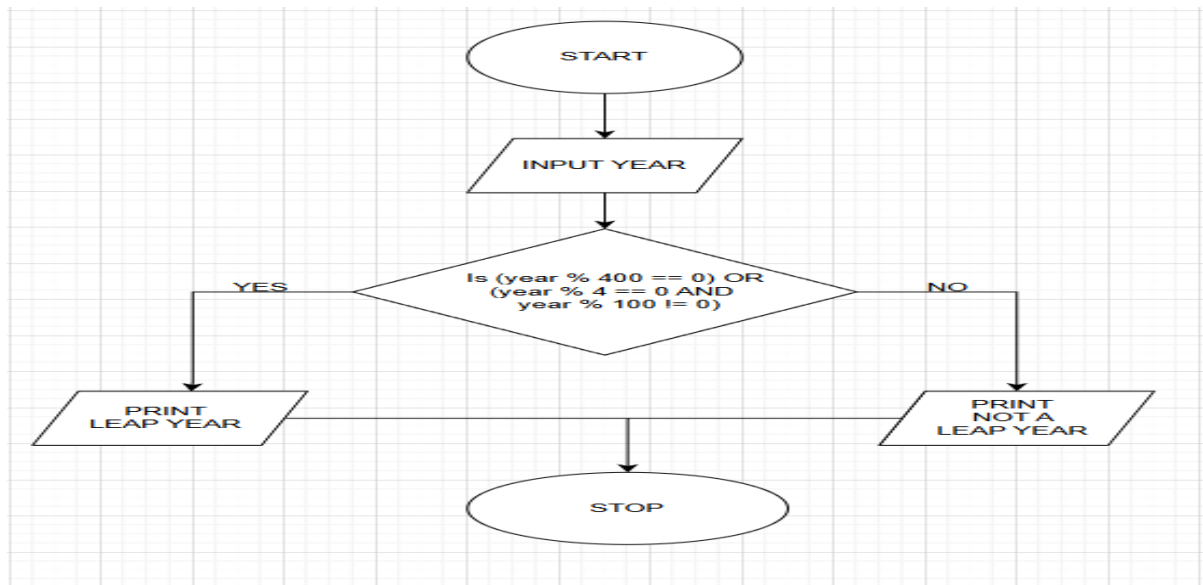
The year is divisible by 400

OR the year is divisible by 4 and not divisible by 100

Step 4: If the condition is true, print "Leap year" Else, print "Not a leap year"

Step 5: Stop

FLOWCHART



PROGRAM

The screenshot displays the CodeTANTRA IDE interface. The left sidebar shows a list of experiments, with '5.1.1 Leap Year Checker' selected. The main editor area contains the following Python code:

```
1 year = int(input())
2 if (year % 4 == 0):
3     if (year % 100 == 0):
4         if (year % 400 == 0):
5             print("Leap year")
6         else:
7             print("Not a leap year")
8     else:
9         print("Leap year")
10 else:
11     print("Not a leap year")
12
```

Below the code editor, the 'Test cases' section shows the results of two test cases:

Test Case	Expected Output	Actual Output	Status
Test case 1	2024	2024	Passed
Test case 2	2023	2023	Passed

The bottom status bar indicates that 2 out of 2 shown test case(s) passed and 2 out of 2 hidden test case(s) passed. The average time for the test cases is 0.005 s, and the maximum time is 0.007 s.

5.1.2 STUDENT GRADES

ALGORITHM

1. Start
2. Input marks of four subjects (m_1, m_2, m_3, m_4)
3. Calculate total
$$total = m_1 + m_2 + m_3 + m_4$$
4. Calculate aggregate percentage
$$aggregate = total/4$$
5. If $aggregate > 75$
→ grade = "Distinction"
6. Else if $aggregate \geq 60$
→ grade = "First Division"
7. Else if $aggregate \geq 50$
→ grade = "Second Division"
8. Else if $aggregate \geq 40$
→ grade = "Third Division"
9. Else
→ grade = "Fail"
10. Print total
11. Print aggregate (two decimal places)
12. Print grade
13. Stop

PROGRAM

5.1.2. Student Grade Based on Aggregate

Write a program to calculate the total marks, aggregate percentage, and grade of a student based on marks in four subjects. The grade is determined as follows:

- Aggregate > 75%: Distinction
- Aggregate >= 60% and < 75%: First Division
- Aggregate >= 50% and < 60%: Second Division
- Aggregate >= 40% and < 50%: Third Division
- Aggregate < 40%: Fail

Input Format:

- Four space-separated integers representing the marks in four subjects.

Output Format:

- The first line should print the total marks.
- The second line should print the aggregate percentage with two decimal places.
- The third line should print the grade.

Constraints:

- 0 <= marks in each subject <= 100

Sample Test Cases

studentG...

```

1 marks = list(map(int, input().split()))
2
3 total_marks = sum(marks)
4 aggregate = (total_marks / 400) * 100
5
6
7
8 if aggregate > 75:
9     grade = "Distinction"
10 elif aggregate >= 60:
11     grade = "First Division"
12 elif aggregate >= 50:
13     grade = "Second Division"
14 elif aggregate >= 40:
15     grade = "Third Division"
16 else:
17     grade = "Fail"
18

```

Average time: 0.007 s, Maximum time: 0.010 s, 7.10 ms, 10.00 ms

5 out of 5 shown test case(s) passed
5 out of 5 hidden test case(s) passed

Test case 1

Expected output	Actual output
85 90 78 88	85 90 78 88
341	341
85.25	85.25
Distinction	Distinction

Terminal Test cases

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